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/**
 * components/ui/confidence-badge.tsx
 *
 * Recommendation #1 – Confidence scores need user education, not just numbers.
 *
 * A plain 87% badge tells a maintenance engineer nothing about whether to
 * pull the engine offline. This component wraps the number in a tooltip that
 * explains what it means for operational decisions, and adds a plain-English
 * label ("Very confident", "Low confidence") so non-technical users can act
 * without reading the percentage at all.
 *
 * Usage:
 *   <ConfidenceBadge confidence={0.87} />
 *   <ConfidenceBadge confidence={prediction.confidence} showPercent />
 */

```

```

import { Badge } from '@components/ui/badge';
import {
  Tooltip,
  TooltipContent,
  TooltipProvider,
  TooltipTrigger,
} from '@components/ui/tooltip';
import { HelpCircle } from 'lucide-react';
import { cn } from '@lib/utils';
import { getConfidenceLabel } from '@lib/severity';

```

// — Props —————

```

interface ConfidenceBadgeProps {
  /** 0-1 float */
  confidence: number;
  /** Show the raw percentage alongside the label */
  showPercent?: boolean;
  size?: 'sm' | 'default';
  className?: string;
  'data-testid'?: string;
}

```

// — Component —————

```

export function ConfidenceBadge({
  confidence,
  showPercent = false,

```

```

    size = 'sm',
    className,
    'data-testid': testId,
  }: ConfidenceBadgeProps) {
    const cfg = getConfidenceLabel(confidence);
    const pct = Math.round(confidence * 100);

    return (
      <TooltipProvider>
        <Tooltip>
          <TooltipTrigger asChild>
            <Badge
              variant={cfg.variant}
              className={cn(
                'cursor-help gap-1',
                size === 'sm' ? 'h-5 text-xs px-1.5' : 'h-6 text-xs px-2',
                className,
              )}
              data-testid={testId}
            >
              {cfg.label}
              {showPercent && (
                <span className="opacity-70"> · {pct}%</span>
              )}
              <HelpCircle className={size === 'sm' ? 'h-3 w-3 opacity-60' : 'h-3.5'} />
            </Badge>
          </TooltipTrigger>
          <TooltipContent className="max-w-xs space-y-1.5">
            <div className="flex items-center justify-between gap-4">
              <p className="font-semibold text-sm">{pct}% confidence</p>
            </div>
            <p className="text-sm text-muted-foreground">{cfg.description}</p>
            {confidence < 0.6 && (
              <p className="text-xs text-amber-500 dark:text-amber-400 font-medium">
                ⚠ Do not schedule maintenance based on this prediction alone.
              </p>
            )}
          </TooltipContent>
        </Tooltip>
      </TooltipProvider>
    );
  }
}

```